# Technical Specification for Portable Household Energy Storage Power Supply

**Portable Household Energy Storage Power Supply**

**Model: UNITY 1 (1+)**

**Version: A/0**

**Customer Name:**

**Customer Confirmation (Seal): No objection to the contents of this specification. Customer Confirmation Date:**

**Technical Specification for Portable Household Energy Storage Power Supply**

Table of Contents

1. Definition of Terms ..... 3

2. Scope of Application ..... 4

3. Normative Reference Documents ..... 4

4. Test Conditions ..... 5

5. Product Introduction ..... 5

6. Functions Introduction ..... 9

7. Configuration List ..... 10

8. Installation and Fixing ..... 10

9. Packaging, Transportation, and Storage Requirements ..... 10

10. Application Conditions ..... 12

11. Precautions ..... 13

12. Other Stipulations ..... 13

# 1. Definition of Terms

**Customer: The purchaser of the Portable Household Energy Storage Power Supply**

**Product: Refer to the product model description: Unless otherwise specified, the product refers to the 1 kWh Portable Household Energy Storage Power Supply purchased by the customer from Ornate Solar, model: UNITY-INFINITY (White/Without base).**

**Cell: The basic unit that realizes the mutual conversion of chemical energy and electrical energy, composed of positive electrode, negative electrode, separator, electrolyte, shell, and terminals, etc..**

**Battery Module: A battery assembly composed of cells connected in series, parallel, or series-parallel connection, with only one pair of positive and negative output terminals, and should also include components such as casing, management, and protection.**

**New Product Status: Refers to the product state within 15 days of customer receipt and with fewer than 5 charge and discharge cycles.**

**Battery Management System (BMS): A device that detects parameters such as battery voltage, current, and temperature information, and manages and controls the battery status.**

**Nominal Voltage: The appropriate approximate voltage value that marks or identifies a type of battery or an electrochemical system.**

**Charge Rate: The ratio of charging power to the energy value of the product measured multiple times by the Battery Management System. For example: when the product energy is 896Wh, and the charging power is 448W, the charge rate is 0.5P; when the product energy decays to 627.2Wh, and the charging power is 313.6W, the charge rate is also 0.5P.**

**Discharge Rate: The ratio of discharge power to the energy value of the product measured multiple times by the Battery Management System. For example: when the product energy is 896Wh, and the discharge power is 448W, the discharge rate is 0.5P; when the product energy decays to 627.2Wh, and the discharge power is 313.6W, the discharge rate is 0.5P.**

**Standard Charge: At room temperature (25±2)°C, charge at a constant power of 0.5P until the voltage of any cell within the module reaches 3.65V, stop charging, and rest for 30min.**

**Standard Discharge: At room temperature (25±2)°C, discharge at a constant power of 0.5P until the voltage of any cell within the module reaches 2.5V, stop discharging, and rest for 30min.**

**Cycle: One cycle is defined as charging and discharging the battery or module once according to the specified standard charge and discharge procedures.**

| Measurement Units |
|---|
| Voltage unit: "V" (Volt) V |
| Current unit: "A" (Ampere) A |
| Power unit: "W" (Watt) W |

| |
|---|
| **Capacity unit: "Ah" (Ampere-Hour) Ah** |
| **Energy unit: "Wh" (Watt-Hour) Wh** |
| **Internal Resistance unit: "mΩ" (milliOhm) mΩ** |
| **Temperature unit: "℃" (degree Celsius) ℃** |
| **Length unit: "mm" (millimeter) mm** |
| **Time unit: "s" (second) s** |
| **Frequency unit: "Hz" (Hertz) Hz** |
| **Mass unit: "kg" (kilogram) kg** |

## 2. Scope of Application

This technical specification stipulates the **detailed introduction of the technical solution, transportation, packaging, storage requirements, and precautions** for the UNITY INFINITY product, among others.

The product specifications, performance, and parameters detailed in this technical specification are for the customer's preliminary reference only. Ornate Solar reserves the right to amend the contents of the technical specification, and the specific terms shall be subject to the contract and documents subsequently signed by both parties.

## 3. Normative Reference Documents

The following documents are indispensable for the application of this document. For dated references, only the cited dated version applies to this document. For undated references, the latest version (including all amendments) applies to this document.

- **UN 38.3** Requirements of the United Nations Manual of Tests and Criteria, Part III, subsection 38.3 – Lithium Batteries
- **IEC 62619-2022** Safety requirements for secondary lithium cells and batteries for use in industrial applications with alkali or other non-acid electrolytes
- **IEC 62620** Performance requirements for secondary lithium cells and batteries for use in industrial applications with alkali or other non-acid electrolytes
- **EN 301489-1, 17** Electromagnetic compatibility (EMC) standard for radio equipment and services
- **EN 300440** Wireless EU CE certification standard, covering electromagnetic compatibility and radio frequency performance for short-range wireless products with a frequency range of 9kHz to 246GHz
- **IEC/EN 61000-6-1, 2, 3** Electromagnetic compatibility (EMC) CE certification standard
- **IEC 60529** Degrees of protection provided by enclosures (IP Code)
- **GB/T 4208-2017** Degrees of protection provided by enclosures (IP Code)
- **GB/T 17626** Electromagnetic compatibility testing and measurement techniques

# 4. Test Conditions

Unless otherwise specified, the product parameters are those in the **new module status**, and the test object is the module in the new module status (except for self-discharge testing). Unless otherwise specified, experiments and measurements shall be conducted at room temperature **(25±2)℃** and standard humidity **(55±20)%**.

## 4.1 Measuring Equipment and Accuracy

(1) Test equipment accuracy: **±0.1%**.

(2) Current measurement accuracy: ≥ **Level 1**.

(3) Voltage measurement accuracy: ≥ **Level 1**.

(4) Temperature measurement accuracy: **±0.5℃**.

(5) Time measurement accuracy: **±0.1%**.

## 4.2 Charge and Discharge Mode

Unless otherwise specified, the charge and discharge modes should all be **standard charge/discharge modes**.

# 5. Product Introduction

The Portable Household Energy Storage Power Supply product mainly consists of **314Ah battery, temperature control system, smart Battery Management System, etc.**. The nominal energy of the **1 kWh Portable Household Energy Storage Power Supply system is 1004.8Wh**, and the nominal energy of the Portable Household Energy Storage Power Supply Combination **(5+1 system) is 5024.0Wh**. It uses **314Ah high-energy density cells**, and the temperature control system is equipped with an **air-cooling circulation system**.

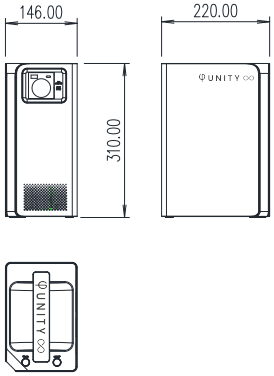
The 1 kWh Portable Household Energy Storage Power Supply product supports multiple charging methods such as **solar panel charging and mains charging**, and integrates multiple interface outputs to meet:

- **Outdoor emergency power supply** (camping, rescue, blackout backup).
- **Mobile device charging** (mobile phones, cameras, laptops).

The Portable Household Energy Storage Power Supply Combination (5+1 system) can also meet energy storage demands at the user side in multi-application scenarios such as **grid support and peak shaving/valley filling**.

---

### (Figure 2: Base) (Figure 3: Combination) (Figure 5: Base)

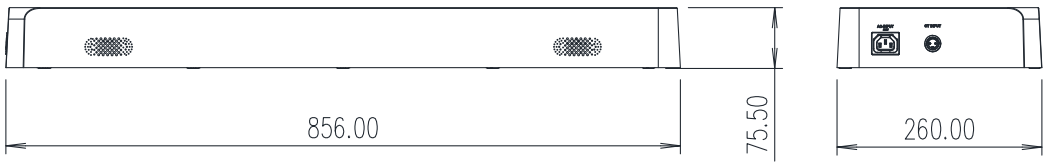


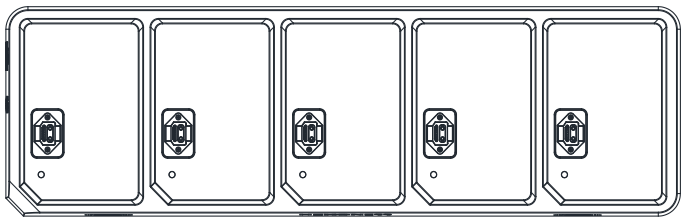
**Portable Household Energy Storage Power Supply Product Model Description:**

**Model Specification: UNITY-①-②-③**

**Model Code for ①: Product Type**

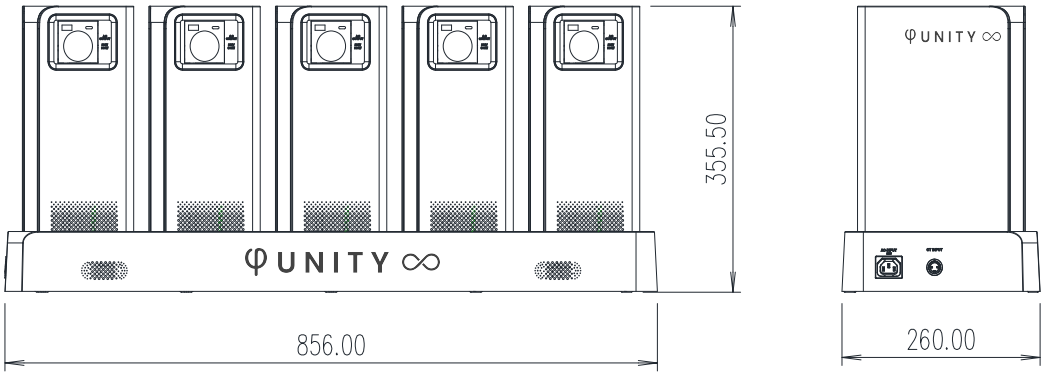
| |
|---|
| • **1** Standard product, battery enclosure color defaults to white (Sky White) |
| • **1+** Customized product, battery enclosure color can be customized upon request |

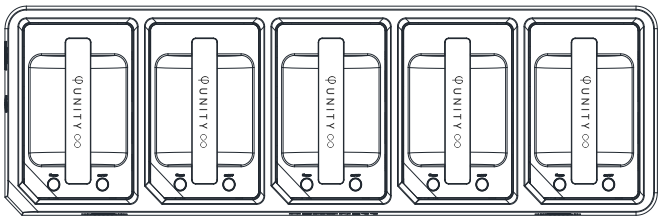


**Model Code for ②: Battery enclosure color and quantity**

| |
|---|
| • **W1**: White; 1: Number of battery enclosures |
| • **G1**: Grey; 1: Number of battery enclosures |
| • **S1**: Silver; 1: Number of battery enclosures |
| • **GD1**: Gold; 1: Number of battery enclosures |



**Model Code for ③: Capacitive touch screen configuration**

- **B**: Base

**Example:** Choosing model **UNITY1-W1**, 1 kWh Portable Household Energy Storage Power Supply, white, 1 battery enclosure, no base configuration.

**Example:** Choosing model **UNITY1-W5-B**, Portable Household Energy Storage Combination Power Supply, white, 5 battery enclosures (300W each) totaling 1500W, configured with a base.

**Example:** Choosing model **UNITY1+-S5-B**, Portable Household Energy Storage Combination Power Supply, silver, 5 battery enclosures (300W each) totaling 1500W, configured with a base.

**The product has the following features:**

1. Uses mature, safe, economical, green, environmentally friendly, ultra-long life, and highly reliable **Lithium Iron Phosphate (LFP) batteries**.
2. Integrated advanced **Battery Management System (BMS)**, providing real-time monitoring and intelligent management, comprehensive battery protection strategy, and fault detection and isolation measures to ensure the safety of the energy storage system.
3. **High-efficiency air-cooling temperature control system**, ensuring cell temperature difference is **<5℃**, smooth temperature fluctuation of the cells, **extending battery life by 20%**.
4. **High-strength, fire-rated plastic enclosure**.
5. **Factory prefabrication manufacturing**, supporting convenient, economical, and efficient deployment in customer households. This can effectively **reduce construction workload by 25%**, increase commissioning and operation/maintenance time by 30%, and **reduce engineering costs by ~10%**.
6. Key components utilize internationally renowned brands, ensuring **reliable quality**.

## 5.1 Main Technical Parameters are shown in the table below

| Item | Parameter | Remarks |
|---|---|---|
| | **Cell** | |
| Cell Type | LFP | |
| Nominal Capacity (Ah) | 314 | 0.5P, 25℃ |
| Nominal Energy (Wh) | **1004.8** | 0.5P, 25℃ |
| Nominal Voltage (V) | 3.2 | |
| | **1 kWh Battery Enclosure** | |
| Configuration | 1P | |
| Nominal Capacity (Ah) | 314 | 0.5P, 25℃ |
| Nominal Energy (kWh) | **1.0048** | 0.5P, 25℃ |
| Nominal Voltage (V) | 220V, 45~55Hz | |
| Operating Voltage (V) | 220±5% | T>0℃ T≤0℃ |
| PV Input Voltage (VDC) | 16~38 | |
| AC Output Maximum Power (W) | AC 220V **300W** | |
| USB Output Power | QC2.0 Quick Charge, **5V 2.1A** | |
| Dimensions (mm) | 146*310*220 | Width * Height * Depth |
| Weight (kg) | ≤10 | |
| Cycle Life | **≥6000 cycles** | 25±10℃, 90%DOD, 80%EOL |
| Cooling Method | Air Cooling | |
| Protection Rating | IP20 | |
| Shipment SOC (%) | 50 | |
| Operating Temperature Range (℃) | -20~60 | Discharge |
| | 0~60 | Charge |
| Storage Temperature (℃) | -20~35 | |
| Installation Method | Indoor Installation/Portable Use | |
| External System Communication | **Bluetooth/WIFI** | |
| Certifications | **CE, UN38.3, ROHS** | |
| | **Base** | |
| Nominal Voltage (V) | 220V, 45~55Hz | |
| Operating Voltage (V) | 220~240 | T>0℃ T≤0℃ |
| Dimensions (mm) | 856* 75.5* 260 | Width * Height * Depth |
| Weight (kg) | ≤5 | |

# 6. Functions Introduction

**6.1 multi-scenario compatibility:** Supports **over 99% of household appliances and digital devices**.

**6.2 Plug and Play:** Appliance-like design, plug-and-play, convenient and fast.

**6.3 Low Noise and No Pollution:** Low noise **<60dB**, zero emissions, environmentally friendly and safe.

**6.4 Smart Management:** Smart Battery Management System, real-time monitoring of voltage, current, and temperature.

**6.5 Parallel Expansion:** Supports parallel capacity expansion of multiple devices (**optional base required**), meeting energy storage demands in application scenarios such as **grid support and peak shaving/valley filling**.

**6.6 APP Remote Control:** Connects via **Bluetooth/WIFI** to mobile phone or tablet APP, allowing real-time viewing of power level and control of switches.

**6.7 Voice Interaction:** (**Optional base required**) Voice query of power level and device status.

**6.8 Replace Only, No Repair: 24-hour response**, shipped from warehouse.

# 7. Configuration List

**Table 7.1 1 kWh Portable Household Energy Storage Power Supply Configuration List**

| No. | Item | Quantity | Remarks |
|---|---|---|---|
| 1 | Battery Enclosure | 1 unit | 1004.8Wh, 146(W)*310(H)*220(D)mm |
| 2 | Power Connection Cable | 1 piece | C14 to French plug, 1.5 square power cable, 3 meters long |
| 3 | User Manual | 1 copy | |
| 4 | Warranty Card | 1 copy | |

**Table 7.2 Portable Household Energy Storage Combination Power Supply Configuration List**

| No. | Item | Quantity | Remarks |
|---|---|---|---|
| 1 | Battery Enclosure | 5 units | 5024.0Wh, Output Voltage AC220V |
| 2 | Power Base | 1 unit | 856(W)*75.5(H)*260(D)mm |
| 3 | Power Connection Cable | 2 pieces | C14 to French plug, 1.5 square power cable, 3 meters long |
| 4 | CT Current Connection Cable | 1 piece | Open-type Current Transformer, 3 meters long |
| 5 | User Manual | 1 copy | |
| 6 | Warranty Card | 1 copy | |

## 8. Installation and Fixing

The storage location should have enough space around it for easy installation and maintenance. Protective preparations should be made in advance, ensuring requirements for **lightning protection, waterproofing, rodent/insect prevention, and ventilation** are met overall.

## 9. Product Identification, Packaging, Transportation, and Storage Requirements

**9.1** Product identification shall be executed according to the standards agreed upon by the customer and UNITY. If there is no agreement, UNITY's standards shall apply.

**9.2** Packaging boxes and packing specifications shall be executed according to the agreement between the customer and UNITY. If there is no agreement, UNITY's standards shall apply.

**9.3** The product shipment report shall include the **capacity, dimensions, and module data within the battery clusters in each container**.

**9.4** When storing the product, it should be placed in a warehouse with good air circulation, relative humidity not exceeding 80%, ambient temperature between **-20℃ and 35℃**. The warehouse must be protected against water, corrosion, and dust. The **SOC should be maintained between 10% and 50%**.

**9.5** Packaging boxes should be marked with phrases such as **"Handle with Care," "Waterproof," "Do Not Invert," and "Stacking Layers Allowed"**.

**9.6** Packaging boxes or cartons must be handled gently during transportation to **avoid collision and knocking**. They must **not be placed together with corrosive substances such as acids and alkalis**.

**9.7** Battery modules should not show adverse conditions such as tilting or collapse during normal consignment or placement.

## 10. Application Conditions

The customer must ensure strict adherence to the following product-related application conditions:

**10.1** The customer must use products matched by batch; **different batches cannot be mixed**.

**10.2** After the customer receives the arrived product, **warehousing inspection should be completed within 15 days**, referencing the inspection specification agreed upon by both parties.

**10.3 Operating temperature range:** Charging: **0～60℃**; Discharging: **-20～60℃**.

**10.4 Short-term storage temperature range (within 1 month): -20～35℃**.

**10.5 Altitude: ≤3000m**.

**10.6 Relative humidity: ≤80%RH**.

**10.7** The system must be configured with a **Battery Management System (BMS)** to closely monitor, manage, and protect every cell.

**10.8** To avoid affecting the product's performance, the customer must **not modify the design and framework of the BMS without authorization**. Otherwise, UNITY shall be exempt from liability for quality issues or accidents that have a direct causal relationship with such modifications.

**10.9** The customer should **maintain complete monitoring data of the internal cell operation** of the product, which will be used as a reference for determining product quality liability. Otherwise, UNITY will not assume product quality assurance responsibility.

**10.10** The customer and third parties must **prevent the product from reaching an over-discharged state**.

- If the voltage of any cell within the module drops **below 2.0V**, the internal parts of the product may suffer **permanent damage**, and UNITY's product quality assurance liability will become invalid.
- When the discharge cut-off voltage of any cell within the module is **lower than 2.5V**, the customer needs to train users to recharge it as soon as possible to prevent the product from entering an over-discharged state.

**10.11 Module storage SOC should be maintained within the range of 20% to 50%**. If the customer anticipates storing the module for **more than 1 month but not exceeding 3 months**, a charge/discharge cycle should be performed beforehand to adjust the SOC to 20% to 50%. UNITY will not be responsible for capacity loss or other losses caused to the module if the storage SOC falls outside the 20% to 50% range or if the module is stored for more than 3 months without charge/discharge maintenance.

# 11. Precautions

**11.1 Do not immerse the product in water**.

**11.2** Improper use and storage of the product pose risks of **fire, explosion, and burns**. Do not disassemble, crush, incinerate, or heat the product.

**11.3 Do not throw the product into fire** or expose it for long periods to high-temperature environments exceeding the temperature conditions specified in this technical specification, as this may lead to fire. Under any normal operating conditions, the cell temperature within the module **must not exceed 60℃**. If the temperature exceeds 60℃, the product must be shut down and operation stopped.

**11.4 Keep the product out of reach of children**. Do not remove the original packaging before use, and dispose of used products promptly in accordance with local recycling or waste regulations.

**11.5 Do not disassemble, dismantle, or repair the product** in any manner without authorization.

**11.6 Do not mix products of different specifications, different brands, or different batches**.

**11.7** If the product emits odor, generates heat, deforms, discolors, or shows any other abnormal phenomena, **stop using it immediately and move it to a safe location**.

**11.8 Short-circuiting the positive and negative terminals of the product is prohibited**, as strong current and high temperatures may cause personal injury or fire. Sufficient safety protection must be in place during the assembly and connection of the battery system to prevent short circuits.

**11.9** Strictly connect the positive and negative terminals of the product according to the labels and instructions. **Reverse or series charging is prohibited**.

**11.10 Overcharging/over-discharging the product is prohibited**, as this may cause overheating of the cells within the module and lead to a fire accident. Multiple hardware and software overcharge and over-discharge fail-safe protections must be implemented during product installation and use.

**11.11 Improper charging termination** may occur during product charging (e.g., termination due to exceeding allowable charging time, charging voltage being too high, or charging current being too strong). When this occurs, it may indicate leakage in the battery system or component failure. Continuing to charge the product before thoroughly resolving the root cause may lead to overheating of the cells or fire.

**11.12** The customer should **securely fix the product onto a solid surface** and safely secure the power cable in an appropriate position to avoid friction leading to arcs and sparks.

**11.13 Electrical connection using plastic is strictly prohibited**. Incorrect electrical connection methods may cause the product to overheat during use.

**11.14** In case of **electrolyte leakage, avoid skin and eye contact** with the electrolyte. If contact occurs, wash the affected area with large amounts of clean water and seek medical attention. It is prohibited for any person or animal to swallow any part of the product.

**11.15** Protective measures must be taken during product use to shield it from **mechanical vibration, collision, and pressure impact**. Otherwise, internal short circuits may occur, generating high temperature and fire.

- The product presents potential hazards, and appropriate protective measures must be taken during operation and maintenance.
- Safety performance tests must only be conducted by **professional personnel equipped with appropriate protective gear in specialized laboratories**. Improper operation during testing may cause cells to catch fire or explode. Non-compliance may cause serious personal injury and property damage.

**11.16** The customer is aware of the potential hazards during operation, including injury from **chemicals, electric shock, or electric arc**. Since DC voltage above **50V** is equally serious as AC voltage in causing harm, the customer must adopt a cautious posture. When operating the product and selecting Personal Protective Equipment (PPE), the customer and employees must consider risks to prevent accidental short circuits leading to electric arcs, explosions, or thermal runaway.

**11.17** The customer should **cease using the product** when its internal resistance **exceeds 200% of its initial internal resistance** or its capacity is **less than or equal to 70% of the nominal capacity**. Otherwise, Ornate Solar shall not be responsible for discrepancies, quality problems, cell failures, or any losses incurred.

# 12. Other Stipulations

**12.1** Ornate Solar can provide services and technical support if required during product installation and use. If product issues arise due to use not in accordance with this specification, Ornate Solar can provide technical guidance but **does not guarantee free replacement service**.

**12.2** The customer must strictly use the product according to this specification and ensure that product users do the same. Otherwise, Ornate Solar shall not be responsible for any product parameter discrepancies, quality issues, failures, or any losses incurred.

**12.3** The customer bears the **obligation of confidentiality** regarding this specification and must not disclose the contents to any third party without authorization. Specific stipulations are detailed in the confidentiality agreement signed by both parties.

**12.4** Without Ornate Solar's written consent, the customer, users, and related parties shall not, under any circumstances, **integrate, separate, or modify the product's technical solution**, nor shall they **probe, reverse engineer, or perform similar actions** on the product.

**12.5 Ornate Solar reserves the right to modify the product specifications and performance parameters**. Customers need to confirm the latest specifications and performance parameters with Ornate Solar before ordering.

**12.6** If product samples are in the development stage, they are **for testing purposes only**. Specific test items must be determined by the customer in consultation with Ornate Solar, and unauthorized sale to any third party is prohibited.

**12.7** If improper use of the product by the customer, product users, or any related party causes social impact and damages UNITY's reputation, they shall **compensate Ornate Solar for all losses**.

**12.8** Acceptance, warranty after-sales, technical services, and other content shall be subject to the **business agreement between both parties**.

**12.9** If there is any inconsistency between this technical specification and other related documents provided by Ornate Solar to the customer, **Ornate Solar reserves the right of final interpretation**.